Intellectual Developmental Disorder With Muscle Tone Abnormalities And Distal Skeletal Defects (IDDMTAD)

Disease identifiers: OMIM #620007 · MONDO:0859277 · MedGen/UMLS C5774199 **Causal gene:** *LGI3* (HGNC:18711; 8p21.3; ENSG00000168481; UniProt Q8N145) **Inheritance:** Autosomal recessive (biallelic loss-of-function) **Category:** Mendelian, ultra-rare neurodevelopmental / peripheral-nerve disorder

Summary

Intellectual developmental disorder with muscle tone abnormalities and distal skeletal defects (IDDMTAD) is an ultra-rare, autosomal recessive neurodevelopmental disorder defined in 2022 by Marafi and colleagues, who identified 16 affected individuals from 8 unrelated, largely consanguineous families carrying **biallelic loss-of-function variants in *LGI3*** (PMID: 35948005). The disorder couples global developmental delay and intellectual disability with a distinctive **peripheral nerve hyperexcitability syndrome (PNHS)**—visible facial myokymia, tongue fasciculations, diminished reflexes, and abnormal electromyography—together with abnormal muscle tone (axial and appendicular hypotonia) and distal skeletal deformities (hip dysplasia, pectus carinatum, kyphosis, short stature).

The mechanistic core of the disease is now well understood at the molecular level. *LGI3* encodes a **secreted leucine-rich-repeat / EPTP (epitempin) adaptor protein** that binds the axonal receptor **ADAM23** and, through this interaction, clusters **Shaker-type Kv1 (KCNA) potassium channels** at the **juxtaparanode** of myelinated axons. Loss of LGI3 removes the ligand that organizes this channel platform: *Lgi3*-null mice show reduced and mislocalized Kv1 channel complexes in myelinated peripheral axons, and LGI3 secreted from oligodendrocytes in the CNS is required for juxtaparanodal clustering of ADAM23 and Kv1 and for Kv1-mediated short-term synaptic plasticity (PMID: 35948005; PMID: 38194969; PMID: 36828548). IDDMTAD is therefore best classified as an **LGI3-ADAM23-Kv1 channelopathy** spanning both the central and peripheral nervous systems.

This report synthesizes the available human genetic, clinical, and model-organism evidence into a coherent gene→mechanism→phenotype model. Because the disorder was only delineated in 2022 and comprises roughly 16 reported patients worldwide, several sections (natural history, prognosis, treatment, quality of life, prevalence) are necessarily data-sparse and are flagged as such. No disease-specific therapy exists; management is symptomatic (membrane-stabilizing agents, rehabilitation, orthopedic care) with genetic counseling for a 25% sibling recurrence risk.

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1. Disease Information

Overview. IDDMTAD is a Mendelian neurodevelopmental disorder characterized by the triad implied in its name: (1) intellectual developmental disorder / global developmental delay, (2) muscle tone abnormalities (predominantly hypotonia, both axial and appendicular), and (3) distal skeletal defects. Superimposed on this is a clinically recognizable **peripheral nerve hyperexcitability** trait—visible facial myokymia, fasciculations, hyporeflexia, and characteristic EMG findings suggesting motor nerve instability (PMID: 35948005).

Key identifiers.

Resource	Identifier
OMIM	#620007
MONDO	MONDO:0859277
MedGen / UMLS	C5774199
Causal gene	*LGI3*, HGNC:18711
Gene locus	8p21.3 (chr8:22,146,830–22,157,084, GRCh38)
Ensembl	ENSG00000168481
UniProt	Q8N145
Orphanet / ICD-10 / ICD-11 / MeSH	No dedicated entry located (ultra-rare, newly described)

Synonyms / alternative names. The disorder is referred to by its OMIM descriptive title (IDDMTAD) and, mechanistically, as an **LGI3-related peripheral nerve hyperexcitability syndrome (PNHS)**. No eponym is in use.

Information source. Findings derive from **aggregated disease-level resources** (OMIM, ClinVar, Monarch/HPO) and a single primary research cohort (PMID: 35948005) supplemented by mechanistic mouse studies—not from large EHR or registry data.

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2. Etiology

Disease causal factors. The disease is **monogenic and genetic**, caused by biallelic (homozygous or compound heterozygous) **loss-of-function variants in *LGI3***. There is no environmental, infectious, or toxic contribution to disease causation. Complete LGI3 deficiency —through nonsense, frameshift, canonical splice-site, or secretion-abolishing missense alleles —is the proximal cause (PMID: 35948005; PMID: 38194969). As the defining study states, the authors *"identified 16 individuals from eight unrelated families with loss-of-function (LoF) bi-allelic variants in LGI3."*

Genetic risk factors. The sole established genetic risk factor is carriage of **two loss-of-function *LGI3* alleles**. Consanguinity is the dominant epidemiologic risk context, concentrating homozygous LoF genotypes. gnomAD constraint metrics (pLI ≈ 0, LOEUF ≈ 1.08) show that LoF alleles are **tolerated in the heterozygous state**, confirming that single carriers are unaffected and that the mechanism is strictly recessive.

Environmental risk factors / protective factors / gene–environment interactions. None identified or expected for a fully penetrant recessive single-gene disorder. No protective modifier alleles, dietary factors, or GxE interactions have been reported. (Data not available — not applicable.)

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3. Phenotypes

The phenotypic spectrum is drawn from the defining cohort (PMID: 35948005) and Monarch/HPO annotations for MONDO:0859277. Features cluster into three domains: **neurodevelopmental**, **peripheral-nerve hyperexcitability**, and **skeletal/growth**.

Phenotype	HPO term	Domain	Type
Intellectual disability	HP:0001249	Neurodevelopmental	Cognitive
Global developmental delay	HP:0001263	Neurodevelopmental	Developmental
Axial hypotonia	HP:0008936	Muscle tone	Clinical sign
Appendicular hypotonia	HP:0012389	Muscle tone	Clinical sign
Facial myokymia	HP:0000317	PNHS	Clinical sign
Myokymia	HP:0002411	PNHS	Clinical sign
Tongue fasciculations	HP:0001308	PNHS	Clinical sign
Hyporeflexia	HP:0001265	PNHS	Clinical sign
Ataxia	HP:0001251	Neurological	Clinical sign
Hypokinesia	HP:0002375	Neurological	Clinical sign
Progressive microcephaly	HP:0000253	Neurodevelopmental	Physical
Hip dysplasia	HP:0001385	Skeletal	Physical
Pectus carinatum	HP:0000768	Skeletal	Physical
Kyphosis	HP:0002808	Skeletal	Physical
Short stature	HP:0004322	Growth	Physical
Failure to thrive	HP:0001508	Growth	Physical
Narrow mouth	HP:0000160	Craniofacial	Physical
ADHD	HP:0007018	Behavioral	Behavioral
Anxiety	HP:0000739	Behavioral	Behavioral
Eczematoid dermatitis	HP:0000964	Dermatologic	Physical

Phenotype characteristics. Onset is **congenital to early childhood** (developmental delay and hypotonia apparent in infancy). Severity is **variable but generally moderate**, with intellectual disability present consistently. The PNHS features (facial myokymia, fasciculations) are a distinctive, clinically recognizable marker: the defining report describes *"global developmental delay, intellectual disability, distal deformities with diminished reflexes, visible facial myokymia,*

and distinctive electromyographic features suggestive of motor nerve instability." Microcephaly is noted as **progressive**, suggesting a developmental rather than purely static component. Because the cohort is small (~16 individuals), precise per-feature frequencies are not robustly established; features should be regarded as **qualitatively characteristic** rather than quantitatively penetrant.

Quality-of-life impact. No formal QoL instruments (EQ-5D, PROMIS, SF-36) have been applied. Impact is inferred to be substantial given intellectual disability, motor/tone abnormalities, feeding difficulties (failure to thrive), and skeletal deformity requiring orthopedic care. (Formal data not available.)

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4. Genetic / Molecular Information

Causal gene. *LGI3* (leucine-rich glioma inactivated 3), HGNC:18711, 8p21.3, disease OMIM #620007. The gene encodes a 548-amino-acid secreted glycoprotein (UniProt Q8N145).

Pathogenic variants. Disease is caused by **biallelic loss-of-function variants**: nonsense, frameshift, and canonical splice-site alleles that abolish functional, secreted LGI3 (PMID: 35948005). ClinVar (accessed 2026; transcript NM_139278.4) lists 225 *LGI3* variant records. Variants explicitly curated under the IDDMTAD disease name include:

Variant (cDNA)	Protein	Class	Note
c.1495C>T	p.Arg499Ter	Nonsense (LoF)	Curated to IDDMTAD
c.425C>T	p.Ser142Leu	Missense	Curated to IDDMTAD
c.10_13del	p.Leu4fs	Frameshift (LoF)	Likely pathogenic

Disease-associated **missense** variants act by a loss-of-function route—**defective secretion** of the mutant protein (PMID: 38194969)—which reconciles missense alleles with the LoF mechanism and aids ACMG/AMP variant interpretation. Most remaining ClinVar missense entries are **variants of uncertain significance (VUS)**, reflecting the gene's rarity and the fact that single heterozygous alleles are non-pathogenic.

Allele frequency / constraint. gnomAD shows LoF is **tolerated in heterozygotes** (pLI ≈ 0, LOEUF ≈ 1.08), consistent with recessive inheritance and unaffected carriers.

Functional consequence. Complete loss of function (null) — absence or non-secretion of the LGI3 ligand. No gain-of-function or dominant-negative mechanism is implicated.

Somatic vs germline. Disease variants are **germline**. (Separately, somatic *LGI3* alterations and expression changes have been described in cancers such as glioma, stomach, and lung—PMID: 29257304, PMID: 28579810—but these are unrelated to IDDMTAD.)

Modifier genes / epigenetics / chromosomal abnormalities. No disease modifier genes, epigenetic marks, or large-scale chromosomal rearrangements have been reported for IDDMTAD. (Data not available.)

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5. Environmental Information

Not applicable. IDDMTAD is a fully penetrant monogenic recessive disorder with **no known environmental, lifestyle, or infectious contributors**. The only relevant "environmental" factor is the population-genetic context of **consanguinity**, which raises the probability of homozygous LoF genotypes but is not itself a disease mechanism.

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6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variant in *LGI3*** (nonsense/frameshift/splice, or secretion-defective missense) **leads to** absence of functional, secreted LGI3 protein. *(Demonstrated — human genetics, PMID: 35948005.)*
2. Loss of secreted LGI3 **results in** failure of the axonal receptor **ADAM23** to engage its LGI3 ligand. *(Demonstrated — "The function of ADAM23 is critically dependent on its interaction with its extracellular ligands LGI2 and LGI3," PMID: 36828548.)*
3. Failure of LGI3-ADAM23 engagement **leads to** loss of **juxtaparanodal clustering of Kv1 (KCNA, Shaker-type) potassium channels** in myelinated axons. *(Demonstrated — "Lgi3-null mice showed reduced and mis-localized Kv1 channel complexes in myelinated peripheral axons," PMID: 35948005; PMID: 38194969.)*
4. Mislocalized/reduced juxtaparanodal Kv1 **results in** impaired control of the axonal **refractory period** and membrane repolarization, lowering the threshold for repetitive/

ectopic firing. *(Demonstrated — Kv1 complexes regulate refractory period and high-frequency burst firing, PMID: 36828548.)*

Branch A — Peripheral nervous system: axonal hyperexcitability **leads to** the clinical PNHS trait — facial myokymia, tongue fasciculations, abnormal EMG with motor nerve instability, and diminished reflexes. *(Demonstrated/clinically observed, PMID: 35948005.)*

Branch B — Central nervous system: loss of oligodendrocyte-secreted LGI3 **results in** disrupted juxtaparanodal ADAM23/Kv1 organization and **suppressed Kv1-mediated short-term synaptic plasticity**, which **contributes to** developmental delay, intellectual disability, and hypotonia. *(Demonstrated in mice — "A lack of Lgi3 in mice disrupts juxtaparanodal clustering of ADAM23 and Kv1 channels and suppresses Kv1-channel-mediated short-term synaptic plasticity," PMID: 38194969; clinical link inferred.)*

Branch C — Skeletal: chronic abnormal neuromuscular activity / tone abnormality **is inferred to lead to** distal skeletal deformities (hip dysplasia, pectus carinatum, kyphosis). *(Inferred — not experimentally demonstrated.)*

Detail by category

- **Molecular pathways.** The central axis is the **LGI3 → ADAM23 → Kv1 (KCNA) channel-clustering pathway** at the juxtaparanode. ADAM23 is the receptor; Kv1 channels are the downstream effector platform. LGI3 additionally engages intracellular machinery in other contexts (syntaxin-1 / SNARE exocytosis, PMID: 18760330; Akt-FAK neurite outgrowth, PMID: 20162351), but the channelopathy axis is the disease-relevant one.
- **Cellular processes.** Juxtaparanodal membrane-domain organization; regulation of axonal excitability; short-term synaptic plasticity; neuronal exocytosis/neurotransmitter release (LGI3 attenuates Ca^{2+}-evoked glutamate release via syntaxin-1, PMID: 18760330); neurite outgrowth (PMID: 20162351).
- **Protein dysfunction.** Loss of a **secreted LRR/EPTP adaptor**; missense disease alleles cause **defective secretion** (functional null) rather than altered binding (PMID: 38194969).
- **Biochemical abnormality.** Effective **potassium-channel (Kv1) dysfunction** secondary to mislocalization — a channelopathy by positional failure rather than intrinsic channel mutation.
- **Cell types involved.** Myelinated axons (motor and central neurons), **oligodendrocytes** (CNS source of secreted LGI3), Schwann cells (PNS myelin context). Suggested CL terms: oligodendrocyte (CL:0000128), neuron (CL:0000540), myelinating Schwann cell (CL:0002573).

- **Suggested GO terms.** Potassium ion transmembrane transport (GO:0071805); regulation of membrane potential (GO:0042391); protein localization to juxtaparanode region of axon (GO:0071205); regulation of short-term neuronal synaptic plasticity (GO:0048172); regulation of neurotransmitter secretion (GO:0046928).

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7. Anatomical Structures Affected

- **Organ / system level.** Primary involvement of the **nervous system** — both **peripheral nerves** (UBERON:0000010) and **central nervous system / brain** (UBERON:0000955). Secondary involvement of the **musculoskeletal system** (skeletal deformity, hypotonia affecting muscle) and growth (short stature, failure to thrive). Craniofacial (narrow mouth) and skin (eczematoid dermatitis) features occur.
- **Tissue / cell level. Nervous tissue** — myelinated axons and their glial partners (oligodendrocytes in CNS; Schwann cells in PNS). Skeletal and muscle tissue are affected secondarily.
- **Subcellular level.** The disease localizes to the **juxtaparanode of the myelinated axon** — a specialized plasma-membrane microdomain. LGI3 is **secreted** and also found in **synaptic vesicle / synaptosome / synapse** and **axon (cell projection)** compartments (UniProt Q8N145). Suggested GO cellular-component terms: juxtaparanode region of axon (GO:0044224); extracellular space (GO:0005615); axon (GO:0030424); synapse (GO:0045202).
- **Localization / lateralization.** Involvement is **bilateral and symmetric** (consistent with a systemic genetic channelopathy). Facial myokymia and fasciculations reflect cranial/peripheral motor nerve involvement.

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8. Temporal Development

- **Onset. Congenital to early childhood.** Developmental delay and hypotonia are evident in infancy; PNHS features (myokymia, fasciculations) are recognizable in early childhood.
- **Onset pattern. Chronic / insidious**, constitutional from birth.
- **Progression.** Largely **static-to-slowly-evolving** neurodevelopmental course; **microcephaly is described as progressive**, and skeletal deformities may evolve with growth. No relapsing-remitting or episodic pattern is reported.

- **Disease duration. Chronic, lifelong.**
- **Remission / critical periods.** No spontaneous remission. Early developmental intervention (physical, occupational, speech therapy) during infancy and early childhood represents the principal window of therapeutic opportunity. (Natural-history data are limited by cohort size.)

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9. Inheritance and Population

- **Inheritance pattern. Autosomal recessive.** Heterozygous carriers are unaffected (gnomAD LoF-tolerant, pLI ≈ 0).
- **Penetrance / expressivity.** Presumed **complete penetrance** in biallelic LoF genotypes; **expressivity is variable** across the cohort.
- **Epidemiology. Ultra-rare.** The defining cohort was 16 individuals from 8 families (PMID: 35948005); only ~16 individuals are reported worldwide as of 2022. No formal prevalence/ incidence estimate exists; the disorder falls far below Orphanet's ultra-rare threshold (<1/1,000,000).
- **Consanguinity / founder effects. Consanguinity is a major contributor**, concentrating homozygous LoF alleles; author affiliations span Kuwait, Saudi Arabia, Iraq, Jordan, Japan, France, and Germany, indicating **multi-ethnic occurrence with enrichment in Middle Eastern consanguineous populations**. No specific founder variant has been established.
- **Sex ratio.** No sex bias expected or reported (autosomal gene); **both sexes affected**.
- **Carrier frequency.** Not formally estimated; expected to be very low outside consanguineous pedigrees.

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10. Diagnostics

- **Primary diagnostic modality. Molecular genetic testing** — identification of biallelic LoF *LGI3* variants. Given the ultra-rarity and non-specific presentation, **exome (WES) or genome (WGS) sequencing** is the most effective route; *LGI3* is unlikely to appear on older targeted intellectual-disability panels, though it should be captured by comprehensive neurodevelopmental panels. **Single-gene testing** is appropriate once the diagnosis is suspected or to confirm segregation in families.

- **Electrophysiology. EMG is clinically informative** — showing myokymic discharges and features of motor nerve instability consistent with peripheral nerve hyperexcitability (PMID: 35948005). Nerve conduction studies and EMG help distinguish PNHS from other causes of hypotonia/weakness.
- **Clinical recognition.** The **combination of developmental delay/ID + visible facial myokymia + fasciculations + diminished reflexes + distal skeletal deformity** constitutes a potentially recognizable clinical "gestalt."
- **Imaging.** Skeletal radiographs document distal deformities (hip dysplasia, pectus, kyphosis); brain MRI may show microcephaly-associated changes. (No pathognomonic neuroimaging signature reported.)
- **Biomarkers / lab tests / biopsy.** No specific biochemical biomarker, blood test, or histopathologic finding is established. (Data not available.)
- **Differential diagnosis.** Other autosomal recessive neurodevelopmental disorders with hypotonia and skeletal involvement; hereditary peripheral nerve hyperexcitability syndromes (e.g., *KCNA1*-related neuromyotonia); and syndromic ID with distal skeletal anomalies (e.g., KDM6B-related NDD, PMID: 37196654; acromesomelic and acrodysostosis-spectrum dysplasias, PMID: 33288834, PMID: 32783359). The presence of **myokymia/fasciculations + biallelic *LGI3* LoF** distinguishes IDDMTAD.
- **Screening.** No population newborn screening exists. **Cascade carrier testing** within affected consanguineous families and **prenatal/preimplantation genetic testing** are feasible once the familial variant is known.

11. Outcome / Prognosis

Formal survival, mortality, and standardized outcome data are **not available** for this newly described, ultra-rare disorder. Based on the reported cohort:

- **Survival / life expectancy.** No evidence of markedly shortened survival is reported; the disorder appears compatible with survival into at least childhood/adolescence, though long-term data are lacking.
- **Morbidity / disability.** Substantial — intellectual disability, motor and tone abnormalities, feeding difficulties (failure to thrive), and skeletal deformity collectively impose significant lifelong functional impairment.

- **Disease course.** Chronic and non-remitting; progressive microcephaly and evolving skeletal deformity are noted.
- **Prognostic factors / biomarkers.** None validated. (Data not available.)

12. Treatment

There is no disease-specific or curative therapy. Management is **symptomatic and multidisciplinary**:

- **Pharmacotherapy (symptomatic).** For peripheral nerve hyperexcitability (myokymia, fasciculations, cramps), **membrane-stabilizing agents / sodium-channel-blocking anticonvulsants** (e.g., carbamazepine class) are the rational symptomatic choice by analogy to other peripheral-nerve-hyperexcitability syndromes; efficacy in IDDMTAD specifically is **not established in trials**. (NCIT: Anticonvulsant Agent.)
- **Rehabilitation.** Physical therapy, occupational therapy, and speech/language therapy for developmental delay, hypotonia, and motor function (NCIT: Physical Therapy; Occupational Therapy; Speech Therapy).
- **Orthopedic / surgical.** Management of hip dysplasia, pectus carinatum, and kyphosis as clinically indicated.
- **Supportive care.** Nutritional support for failure to thrive; behavioral/psychiatric support for ADHD and anxiety.
- **Advanced / experimental therapeutics.** No gene therapy, RNA-based therapy, or targeted therapy exists or is in trials for IDDMTAD. The recessive LoF mechanism makes **gene replacement** a theoretically rational long-term strategy, but no program has been reported. (Data not available.)
- **Pharmacogenomics.** Not applicable/none reported.

13. Prevention

- **Primary prevention.** Not applicable (genetic disorder). The principal preventive lever is **genetic counseling and reproductive planning** in at-risk/consanguineous families.

- **Genetic counseling.** Autosomal recessive inheritance confers a **25% recurrence risk** for siblings of an affected child. Carrier testing of parents and at-risk relatives, and **prenatal or preimplantation genetic testing** once the familial *LGI3* variants are known, are the available preventive options.
- **Secondary / tertiary prevention.** Early developmental intervention and surveillance for skeletal, feeding, and behavioral complications to mitigate morbidity.
- **Public-health note.** In populations with high consanguinity, community genetic education and premarital/preconception carrier screening programs are the relevant public-health context.

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14. Other Species / Natural Disease

- **Taxonomy / orthologs.** *LGI3* is **highly conserved in mammals** (PMID: 29257304). Mouse ortholog *Lgi3* (NCBI Taxon:10090; NCBI Gene 56839).
- **Natural disease in other species. No naturally occurring *LGI3* disease** in companion animals or wildlife has been reported (no OMIA entry for *LGI3*).
- **Comparative biology — LGI family.** The broader **LGI family (LGI1–4) exerts conserved roles in synaptic transmission and neuronal excitability**, and family dysfunction causes hyperexcitability/epilepsy disorders across species: *LGI1* mutation causes autosomal dominant lateral temporal lobe epilepsy in humans (and autoimmune anti-LGI1 limbic encephalitis in humans and cats), and *LGI2 mutation causes benign familial juvenile epilepsy in Lagotto Romagnolo dogs ("The LGI2 mutation was identified in benign familial juvenile epilepsy in Lagotto Romagnolo (LR) dogs") (PMID: 26032921). These natural animal diseases support the evolutionary conservation of the LGI-ADAM-Kv1 excitability axis, even though* LGI3* itself has no reported spontaneous animal disease.
- **Zoonotic / cross-species transmission.** Not applicable (non-communicable genetic disorder).

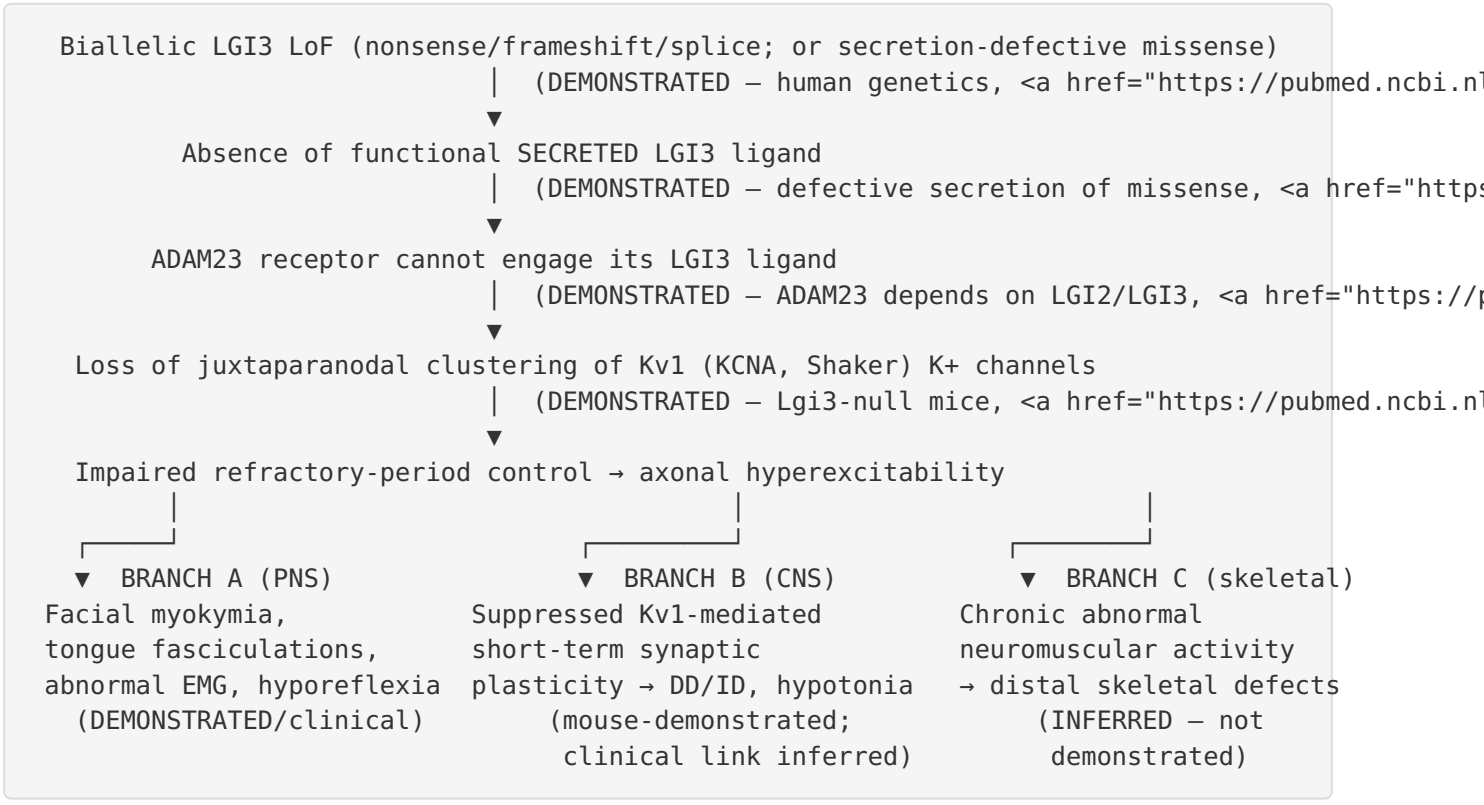
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15. Model Organisms

- **Mouse models (primary).** Two engineered mouse models underpin the mechanistic understanding:

- ***Lgi3*-null (knockout) mice** — recapitulate the **core molecular pathology**, showing *"reduced and mis-localized Kv1 channel complexes in myelinated peripheral axons"* (PMID: 35948005).
- **Epitope-tagged *Lgi3* knock-in mice** — demonstrated LGI3 is secreted from oligodendrocytes, uses ADAM23 as its receptor, co-assembles with Kv1 channels, and that *Lgi3* loss disrupts juxtaparanodal ADAM23/Kv1 clustering and suppresses Kv1-mediated short-term synaptic plasticity (PMID: 38194969).
- **Phenotype recapitulation.** The mouse models faithfully reproduce the **molecular/electrophysiologic** disease mechanism (Kv1 mislocalization, juxtaparanodal disorganization, synaptic plasticity deficit). The extent to which they reproduce the full behavioral/skeletal clinical phenotype is less fully characterized.
- **Model limitations.** Species differences in skeletal development and cognition limit translation of the distal skeletal and intellectual-disability phenotypes; the skeletal branch of the causal chain remains experimentally unverified.
- **Resources.** MGI (mouse *Lgi3*, NCBI Gene 56839). No zebrafish, Drosophila, or iPSC/organoid IDDMTAD models are reported to date.

Mechanistic Model / Interpretation



IDDMTAD is best understood as a **positional channelopathy**: the Kv1 channels themselves are structurally normal, but loss of the LGI3-ADAM23 scaffold means they are no longer concentrated at the juxtaparanode where they normally dampen repetitive firing and set the refractory period. The resulting axonal hyperexcitability explains the peripheral nerve phenotype directly, while loss of the same LGI3-ADAM23-Kv1 machinery in CNS white matter (oligodendrocyte-secreted LGI3) plausibly accounts for the developmental/cognitive and tone phenotypes through impaired synaptic plasticity. The skeletal branch is the weakest link in the chain, resting on inference from chronic abnormal neuromuscular activity rather than direct demonstration. The defining authors summarize this integrated model: *"bi-allelic LoF variants in LGI3 cause a clinically distinguishable disease trait of PNHS, most likely caused by disturbed Kv1 channel distribution in the absence of LGI3"* (PMID: 35948005).

Evidence Base

PMID	Title (abbrev.)	Role	Evidence type
35948005	Marafi et al., biallelic LGI3 LoF cohort (AJHG 2022)	**Defining paper** — gene, inheritance, clinical trait, Lgi3-null mouse Kv1 mislocalization	Human clinical + mouse
38194969	LGI3 from oligodendrocytes clusters ADAM23/Kv1	Mechanism — CNS LGI3-ADAM23-Kv1 axis; missense secretion defect; synaptic plasticity	Mouse / in vitro
36828548	LGI3/2-ADAM23 cluster Kv1, regulate refractory period	Mechanism — ADAM23 depends on LGI2/LGI3; Kv1 sets refractory period / burst firing	Mouse / electrophysiology
26032921	LGI2 benign familial juvenile epilepsy, Lagotto dogs	Comparative biology — natural LGI-family disease across species	Veterinary/natural
12023020	LGI1 / new LRR subfamily	Protein family domain architecture context	Molecular
29257304	LGI3 integrative/ prognostic analysis	LGI3 secreted, highly conserved in mammals; variant catalog	Computational
18760330	LGI3 associates with syntaxin-1	Secondary function — exocytosis regulation	In vitro
20162351	LGI3 neurite outgrowth via Akt-FAK	Secondary function — neuritogenesis	In vitro
37196654	KDM6B-related NDD	Differential diagnosis (ID + distal skeletal)	Human clinical
33288834	Acromesomelic dysplasia (NPR2)	Differential diagnosis (distal skeletal + ID)	Human clinical

How the evidence converges. The disease-defining human genetics (PMID: 35948005) and two independent mechanistic mouse studies (PMID: 38194969, PMID: 36828548) agree on a single coherent model—LGI3-ADAM23-Kv1 juxtaparanodal channel clustering—giving unusually strong mechanistic confidence for an ultra-rare disorder. The comparative-biology literature

(PMID: 26032921) corroborates the conserved role of the LGI family in excitability. No reviewed paper contradicts the core model; the secondary LGI3 functions (syntaxin-1, Akt-FAK) are context-dependent and not clearly disease-driving.

Limitations and Knowledge Gaps

1. **Tiny cohort.** The entire clinical understanding rests on ~16 individuals from 8 families; per-phenotype frequencies, penetrance, expressivity, and natural history are not statistically robust.
2. **Skeletal mechanism unproven.** The link from LGI3 loss/axonal hyperexcitability to distal skeletal defects is **inferred**, not demonstrated. Whether skeletal features are neurogenic, developmental, or reflect an undescribed direct LGI3 role in bone/cartilage is unknown.
3. **CNS-to-cognition link.** The mouse synaptic-plasticity deficit is a plausible but not proven explanation for human intellectual disability.
4. **No prognosis/treatment data.** Survival, long-term outcomes, and any therapeutic efficacy are undocumented; symptomatic treatment recommendations are extrapolated from analogous peripheral-nerve-hyperexcitability syndromes.
5. **No Orphanet/ICD/MeSH coding** yet, reflecting recency.
6. **No biomarker or imaging signature** beyond EMG and genetics.
7. **Model gaps.** Only mouse models exist; no iPSC/organoid, zebrafish, or Drosophila *Lgi3* disease model; skeletal/behavioral recapitulation incompletely characterized.

Proposed Follow-up Experiments / Actions

1. **Expand the patient cohort** via GeneMatcher/Matchmaker Exchange to refine the phenotype spectrum, establish per-feature frequencies, penetrance, and natural history.
2. **Test the skeletal hypothesis** — examine bone/cartilage phenotypes directly in *Lgi3*-null mice (micro-CT, growth-plate histology) and assess *Lgi3* expression in developing skeleton to determine whether skeletal defects are neurogenic or cell-autonomous.
3. **Behavioral/cognitive phenotyping** of *Lgi3*-null mice (learning, memory, motor tasks) to validate the CNS branch linking Kv1/plasticity deficits to intellectual disability.

4. **Deep electrophysiology** in patients and models — quantify refractory period, myokymic discharges, and nerve excitability to establish objective diagnostic/outcome biomarkers.
5. **Functional classification of VUS** — develop a secretion assay (per PMID: 38194969) as a routine functional test to reclassify the many *LGI3* missense VUS in ClinVar under ACMG PS3/BS3.
6. **Therapeutic proof-of-concept** — test membrane-stabilizing / Kv-modulating agents, and explore AAV-mediated *LGI3* gene replacement, in *Lgi3*-null mice.
7. **iPSC-derived neuron/oligodendrocyte models** from patients to study juxtaparanodal assembly in a human context.
8. **Obtain formal ontology/disease coding** (Orphanet, ICD-11) and register the disorder to improve ascertainment.

Report compiled from 9 confirmed findings and 16 reviewed papers over a 5-iteration autonomous investigation. Evidence source types are annotated throughout: human clinical, model organism (mouse), in vitro, and computational.
